Homework

❑ 7.1, 7.5, 7.19, 7.24

7.1 Determine the resultant of the superposition of the parallel waves $E_1 = E_{01} \sin(\omega t + \varepsilon_1)$ and $E_2 = E_{02} \sin(\omega t + \varepsilon_2)$ when $\omega = 120\pi$, $E_{01} = 6$, $E_{02} = 8$, $\varepsilon_1 = 0$, and $\varepsilon_2 = \pi\ 2$. Plot each function and the resultant.

7.1 $E_0^2 = 36 + 64 + 2 \cdot 6 \cdot 8 \cos \pi/2 = 100$, $E_0 = 10$; $\tan \alpha = \frac{8}{6}$, $\alpha = 53.1° = 0.93$ rad.

$$E = 10 \sin(120\pi t + 0.93)$$

7.5 Answer the following:

(a) How many wavelengths of $\lambda_0 = 500$ nm light will span a 1-m gap in vacuum?

(b) How many waves span the gap when a glass plate 5 cm thick ($n = 1.5$) is inserted in the path?

(c) Determine the *OPD* between the two situations.

(d) Verify that Λ/λ_0 corresponds to the difference between the solutions to (a) and (b) above.

7.5 $\dfrac{1 \text{ m}}{500 \text{ nm}} = 0.2 \times 10^7 = 2\,000\,000$ waves

$$\text{In the glass } \frac{0.05}{\lambda_0/n} = \frac{0.05(1.5)}{500 \text{ nm}} = 1.5 \times 10^5$$

$$\text{in air } \frac{0.95}{\lambda_0} = 0.19 \times 10^7$$

total 2 050 000 waves.

$$OPD = [(1.5)(0.05) + (1)(0.95)] - (1)(1)$$
$$OPD = 1.025 - 1.000 = 0.025 \text{ m}$$

$$\frac{\Lambda}{\lambda_0} = \frac{0.025}{500 \text{ nm}} = 5 \times 10^4 \text{ waves}$$

7.19 Given the dispersion relation $\omega = ak^2$, compute both the phase and group velocities.

7.19 $v = \omega/k = ak, \qquad v_g = d\omega/dk = 2ak = 2v$

7.24 Show that the group velocity can be written as

$$v_g = \frac{c}{n + \omega(dn/d\omega)}$$

7.24 $v_g = v + k\dfrac{dv}{dk}$ and $\dfrac{dv}{dk} = \dfrac{dv}{d\omega}\dfrac{d\omega}{dk} = v_g\dfrac{dv}{d\omega}$

Since $v = c/n$, $\dfrac{dv}{d\omega} = \dfrac{dv}{dn}\dfrac{dn}{d\omega} = -\dfrac{c}{n^2}\dfrac{dn}{d\omega}$

$$v_g = v - \frac{v_g ck}{n^2}\frac{dn}{d\omega} = \frac{v}{1 + (ck/n^2)(dn/d\omega)} = \frac{c}{n + \omega(dn/d\omega)}$$

Homework

- ❑7.32, 7.33, 7.37, 7.39

In Problem 7.33, the figure 7.20 should be 7.30

7.32 Show that

$$\int_0^\lambda \sin akx \cos bkx\, dx = 0 \qquad [7.44]$$

$$\int_0^\lambda \cos akx \cos bkx\, dx = \frac{\lambda}{2}\delta_{ab} \qquad [7.45]$$

$$\int_0^\lambda \sin akx \sin bkx\, dx = \frac{\lambda}{2}\delta_{ab} \qquad [7.46]$$

where $a \neq 0$, $b \neq 0$, and a and b are positive integers.

7.32 $\int_0^\lambda \sin akx \sin bkx\, dx$

$$= \frac{1}{2k}\left[\int_0^\lambda \cos[(a-b)kx]k\, dx - \int_0^\lambda \cos[(a+b)kx]k\, dx\right]$$

$$= \frac{1}{2k}\left.\frac{\sin(a-b)kx}{a-b}\right|_0^\lambda - \frac{1}{2k}\left.\frac{\sin(a+b)kx}{a+b}\right|_0^\lambda$$

$$= 0 \quad \text{if } a \neq b$$

Whereas if $a = b$

$$\int_0^\lambda \sin^2 akx\, dx = \frac{1}{2k}\int_0^\lambda (1 + \cos 2akx)k\, dx = \frac{\lambda}{2}$$

The other integrals are similar.

7.33 Compute the Fourier series components for the periodic function shown in Fig. 7.20.

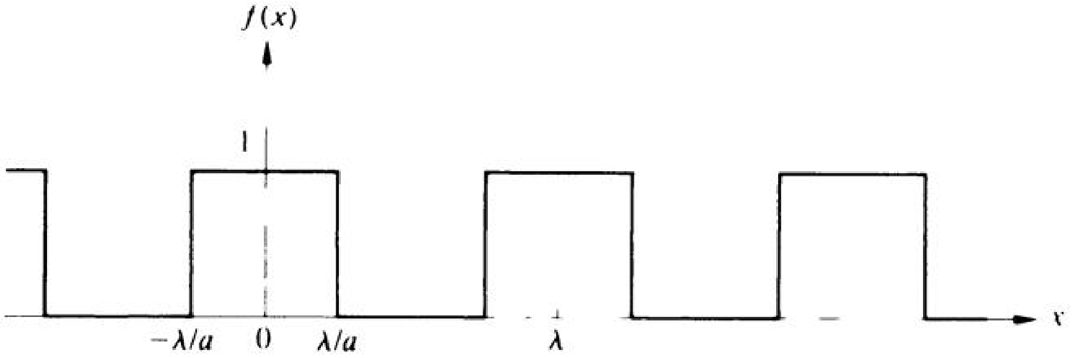


Figure 7.30 A periodic anharmonic function.

7.33 Even function. therefore $B_m = 0$.

$$A_0 = \frac{2}{\lambda}\int_{-\lambda/a}^{\lambda/a} dx = \frac{2}{\lambda}\left(\frac{\lambda}{a} + \frac{\lambda}{a}\right) = \frac{4}{a}$$

$$A_m = \frac{2}{\lambda}\int_{-\lambda/a}^{\lambda/a} (1)\cos mkx\, dx$$

$$A_m = \frac{2}{mk\lambda}\sin mkx\Bigg]_{-\lambda/a}^{\lambda/a}$$

$$A_m = \frac{2}{m\pi}\sin\frac{m2\pi}{a}$$

7.37 Change the upper limit of Eq. (7.59) from ∞ to a and evaluate the integral. Leave the answer in terms of the so-called *sine integral*:

$$\mathrm{Si}(z) = \int_0^z \operatorname{sinc} w \, dw$$

which is a function whose values are commonly tabulated.

7.37
$$f'(x) = \frac{1}{\pi}\int_0^a E_0 L \frac{\sin kL/2}{kL/2} \cos kx \, dk$$
$$= \frac{E_0 L}{\pi 2}\int_0^b \frac{\sin (kL/2 + kx)}{kL/2} dk$$
$$+ \frac{E_0 L}{\pi 2}\int_0^b \frac{\sin (kL/2 - kx)}{kL/2} dk$$

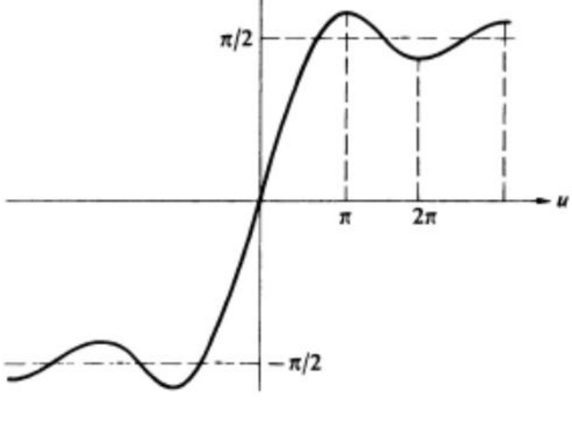
Let $kL/2 = w$, $(L/2)\,dk = dw$, $kx = wx'$.

$$f'(x) = \frac{E_0}{\pi}\int_0^b \frac{\sin(w + wx')}{w} dw + \frac{E_0}{\pi}\int_0^b \frac{\sin(w - wx')}{w} dw$$

where $b = aL/2$. Let $w + wx' = t$, $dw/w = dt/t$. $0 \le w \le b$ and $0 \le t \le (x' + 1)b$. Let $w - wx' = -t$ in the other integral. $0 \le w \le b$ and $0 \le t \le (x' - 1)b$.

$$f'(x) = \frac{E_0}{\pi}\int_0^{(x'+1)b} \frac{\sin t}{t} dt - \frac{E_0}{\pi}\int_0^{(x'-1)b} \frac{\sin t}{t} dt$$

$$f'(x) = \frac{E_0}{\pi}\,\mathrm{Si}\,[b(x' + 1)] - \frac{E_0}{\pi}\,\mathrm{Si}\,[b(x' - 1)], \qquad x' = 2x/L$$



7.39 Derive an expression for the coherence length (in vacuum) of a wavetrain that has a frequency bandwidth $\Delta\nu$; express your answer in terms of the *linewidth* $\Delta\lambda_0$ and the mean wavelength $\bar{\lambda}_0$ of the train.

7.39 $\Delta l_c = c\ \Delta t_c$, $\Delta l_c \approx c/\Delta\nu$. But $\Delta\omega/\Delta k_0 = \bar{\omega}/\bar{k}_0 = c$; thus $|\Delta\nu/\Delta\lambda_0| = \bar{\nu}/\bar{\lambda}_0$,

$$\Delta l_c \approx \frac{c\bar{\lambda}_0}{\Delta\lambda_0\bar{\nu}} \qquad \Delta l_c \approx \bar{\lambda}_0^2/\Delta\lambda_0$$

Or try using the uncertainty principle:

$$\Delta l \approx \frac{h}{\Delta p} \quad \text{where } p = h/\lambda \text{ and } \Delta\lambda_0 << \bar{\lambda}_0$$

Homework

❑8.1, 8.3, 8.21

8.1 Describe completely the state of polarization of each of the following waves:

(a) $\vec{\mathbf{E}} = \hat{\mathbf{i}}E_0 \cos(kz - \omega t) - \hat{\mathbf{j}}E_0 \cos(kz - \omega t)$

(b) $\vec{\mathbf{E}} = \hat{\mathbf{i}}E_0 \sin 2\pi(z/\lambda - \nu t) - \hat{\mathbf{j}}E_0 \sin 2\pi(z/\lambda - \nu t)$

(c) $\vec{\mathbf{E}} = \hat{\mathbf{i}}E_0 \sin(\omega t - kz) + \hat{\mathbf{j}}E_0 \sin(\omega t - kz - \pi/4)$

(d) $\vec{\mathbf{E}} = \hat{\mathbf{i}}E_0 \cos(\omega t - kz) + \hat{\mathbf{j}}E_0 \cos(\omega t - kz + \pi/2)$.

8.1

(a) $\vec{\mathbf{E}} = \hat{\mathbf{i}}E_0 \cos(kz - \omega t) + \hat{\mathbf{j}}E_0 \cos(kz - \omega t + \pi)$. Equal amplitudes, E_y lags E_x by π. Therefore $\mathcal{P}$-state at 135° or −45°.

(b) $\vec{\mathbf{E}} = \hat{\mathbf{i}}E_0 \cos(kz - \omega t - \pi/2) + \hat{\mathbf{j}}E_0 \cos(kz - \omega t + \pi/2)$. Equal amplitudes, E_y lags E_x by π. Therefore same as (a).

(c) E_x leads E_y by $\pi/4$. They have equal amplitudes. Therefore it is an ellipse tilted at +45° and is left-handed.

(d) E_y leads E_x by $\pi/2$. They have equal amplitudes. Therefore it is an $\mathcal{R}$-state.

8.3 Analytically, show that the superposition of an $\mathcal{R}$- and an $\mathcal{L}$-state having different amplitudes will yield an $\mathcal{E}$-state, as shown in Fig. 8.8. What must ε be to duplicate that figure?

8.3 $\vec{\mathbf{E}}_{\mathcal{R}} = \hat{\mathbf{i}}E_0 \cos(kz - \omega t) + \hat{\mathbf{j}}E_0 \sin(kz - \omega t)$

$$\vec{\mathbf{E}}_{\mathcal{L}} = \hat{\mathbf{i}}E_0' \cos(kz - \omega t) - \hat{\mathbf{j}}E_0' \sin(kz - \omega t)$$

$$\vec{\mathbf{E}} = \vec{\mathbf{E}}_{\mathcal{R}} + \vec{\mathbf{E}}_{\mathcal{L}} = \hat{\mathbf{i}}(E_0 + E_0') \cos(kz - \omega t)$$

$$+ \hat{\mathbf{j}}(E_0 - E_0') \sin(kz - \omega t).$$

Let $E_0 + E_0' = E_{0x}''$ and $E_0 - E_0' = E_{0y}''$; then $\vec{\mathbf{E}} = \hat{\mathbf{i}}E_{0x}'' \cos(kz - \omega t) + \hat{\mathbf{j}}E_{0y}'' \sin(kz - \omega t)$. From Eqs. (8.11) and (8.12) it is clear that we have an ellipse where $\varepsilon = -\pi/2$ and $\alpha = 0$.

8.21 An ideal polarizer is rotated at a rate ω between a similar pair of stationary crossed polarizers. Show that the emergent flux density will be modulated at four times the rotational frequency. In other words, show that

$$I = \frac{I_1}{8}(1 - \cos 4\omega t)$$

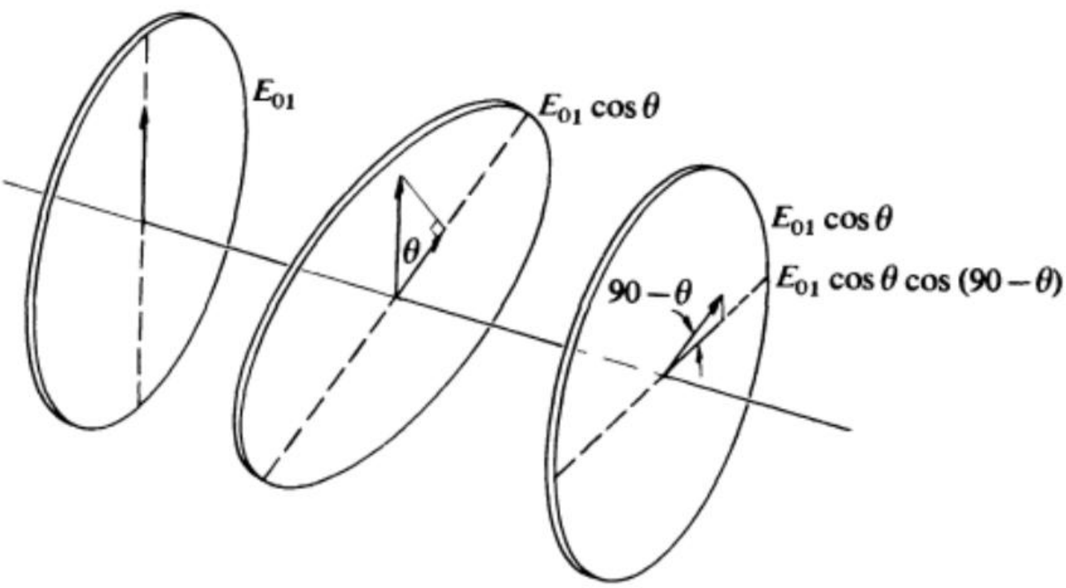
where I_1 is the flux density emerging from the first polarizer and I is the final flux density.

8.21 From the figure, it follows that

$$I = \tfrac{1}{2}E_{01}^2 \sin^2\theta \cos^2\theta = \frac{E_{01}^2}{8}(1 - \cos 2\theta)(1 + \cos 2\theta)$$

$$= \frac{E_{01}^2}{8}(1 - \cos^2 2\theta) = \frac{E_{01}^2}{8}[1 - (\tfrac{1}{2}\cos 4\theta + \tfrac{1}{2})]$$

$$= \frac{E_{01}^2}{16}(1 - \cos 4\theta) = \frac{I_1}{8}(1 - \cos 4\theta) \qquad \theta = \omega t$$



Homework

❑ 8.40

8.40 The prism shown in Fig. P.8.40 is known as a *Rochon polarizer*. Sketch all the pertinent rays, assuming

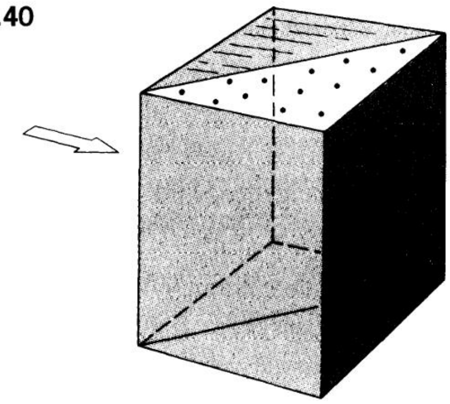
(a) that it is made of calcite.

(b) that it is made of quartz.

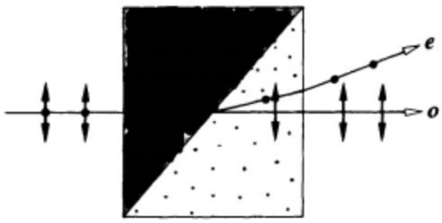
(c) Why might such a device be more useful than a dichroic polarizer when functioning with high–flux density laser light?

(d) What valuable feature of the Rochon is lacking in the Wollaston polarizer?

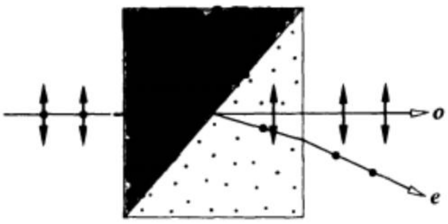
Figure P.8.40



8.40



(a) Calcite



(b) Quartz

(c) Undesired energy in the form of one of the $\mathcal{P}$-states can be disposed of without local heating problems.

(d) The Rochon transmits an undeviated beam (the *o*-ray), which is therefore achromatic as well.

Homework

❑ 8.45, 8.46, 8.49

8.45 A Babinet compensator is positioned at 45° between crossed linear polarizers and is being illuminated with sodium light. When a thin sheet of mica (indices 1.599 and 1.594) is placed on the compensator, the black bands all shift by $\frac{1}{4}$ of the space separating them. Compute the retardance of the sheet and its thickness.

8.45

$$\Delta\varphi = \frac{2\pi}{\lambda_0} d\,\Delta n$$

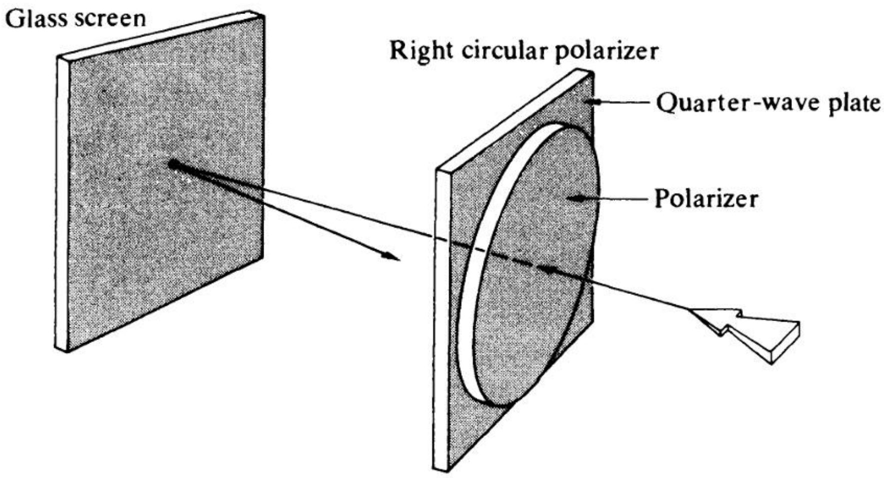
but $\Delta\varphi = (1/4)(2\pi)$ because of the fringe shift. Therefore $\Delta\varphi = \pi/2$ and

$$\frac{\pi}{2} = \frac{2\pi\, d\,(0.005)}{589.3 \times 10^{-9}}$$

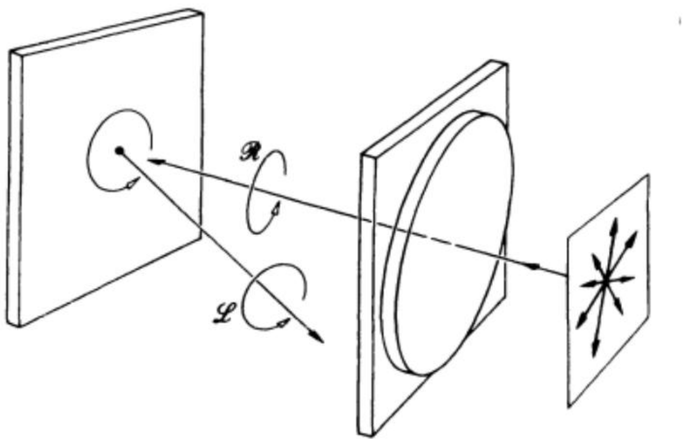
$$d = \frac{589.3 \times 10^{-9}}{2(10^{-2})} = 2.94 \times 10^{-5}\ \mathrm{m}$$

8.46 Imagine that we have randomly polarized room light incident almost normally on the glass surface of a radar screen. A portion of it would be specularly reflected back toward the viewer and would thus tend to obscure the display. Suppose now that we cover the screen with a right-circular polarizer, as shown in Fig. P.8.46. Trace the incident and reflected beams, indicating their polarization states. What happens to the reflected beam?

Figure P.8.46



8.46 The $\mathcal{R}$-state incident on the glass screen drives the electrons in circular orbits, and they reradiate reflected circular light whose $\vec{\mathbf{E}}$-field rotates in the same direction as that of the incoming beam. But the propagation direction has been reversed on reflection, so that although the incident light is in an $\mathcal{R}$-state, the reflected light is left-handed. It will therefore be completely absorbed by the right-circular polarizer. This is illustrated in the figure below.



8.49 On examining a piece of stressed photoelastic material between crossed linear polarizers, we would see a set of colored bands (isochromatics) and, superimposed on these, a set of dark bands (isoclinics). How might we remove the isoclinics, leaving only the isochromatics? Explain your solution. Incidentally, the proper arrangement is independent of the orientation of the photoelastic sample.

8.49 Place the photoelastic material between circular polarizers with both retarders facing it (as in Fig. 8.51). Under circular illumination no orientation of the stress axes is preferred over any other, and they will thus all be indistinguishable. Only the birefringence will have an effect, and so the isochromatics will be visible. If the two polarizers are different, that is, one an $\mathcal{R}$, the other an $\mathcal{L}$, regions where Δn leads to $\Delta\varphi = \pi$ will appear bright. If they are the same, such regions appear dark.

Homework

- 9.1, 9.2, 9.4, 9.26, 9.29

9.1 Returning to Section 9.1, let

$$\tilde{\mathbf{E}}_1(\vec{r}, t) = \tilde{\mathbf{E}}_1(\vec{r})e^{-i\omega t}$$

and

$$\tilde{\mathbf{E}}_2(\vec{r}, t) = \tilde{\mathbf{E}}_2(\vec{r})e^{-i\omega t}$$

where the wavefront shapes are not explicitly specified, and $\tilde{\mathbf{E}}_1$ and $\tilde{\mathbf{E}}_2$ are complex vectors depending on space and initial phase angle. Show that the interference term is then given by

$$I_{12} = \tfrac{1}{2}(\tilde{\mathbf{E}}_1 \cdot \tilde{\mathbf{E}}_2^* + \tilde{\mathbf{E}}_1^* \cdot \tilde{\mathbf{E}}_2) \qquad (9.109)$$

You will have to evaluate terms of the form

$$\langle \tilde{\mathbf{E}}_1 \cdot \tilde{\mathbf{E}}_2 e^{-2i\omega t} \rangle_{\mathrm{T}} = (\tilde{\mathbf{E}}_1 \cdot \tilde{\mathbf{E}}_2 / T) \int_t^{t+T} e^{-2i\omega t'} dt'$$

for $T >> \tau$ (take another look at Problem 3.10). Show that Eq. (9.109) leads to Eq. (9.11) for plane waves.

9.1 $\vec{\mathbf{E}}_1 \cdot \vec{\mathbf{E}}_2 = \frac{1}{2}(\vec{\mathbf{E}}_1 e^{-i\omega t} + \vec{\mathbf{E}}_1^* e^{i\omega t}) \cdot \frac{1}{2}(\vec{\mathbf{E}}_2 e^{-i\omega t} + \vec{\mathbf{E}}_2^* e^{i\omega t})$,

where $\mathrm{Re}\,(z) = \frac{1}{2}(z + z^*)$.

$$\vec{\mathbf{E}}_1 \cdot \vec{\mathbf{E}}_2 = \tfrac{1}{4}[\vec{\mathbf{E}}_1 \cdot \vec{\mathbf{E}}_2 e^{-2i\omega t} + \vec{\mathbf{E}}_1^* \cdot \vec{\mathbf{E}}_2^* e^{2i\omega t} + \vec{\mathbf{E}}_1 \cdot \vec{\mathbf{E}}_2^* + \vec{\mathbf{E}}_1^* \cdot \vec{\mathbf{E}}_2]$$

The last two terms are time independent, while

$$\langle \vec{\mathbf{E}}_1 \cdot \vec{\mathbf{E}}_2 e^{-2i\omega t} \rangle \to 0 \quad \text{and} \quad \langle \vec{\mathbf{E}}_1^* \cdot \vec{\mathbf{E}}_2^* e^{2i\omega t} \rangle \to 0$$

because of the $1/T\omega$ coefficient. Thus

$$I_{12} = 2\langle \vec{\mathbf{E}}_1 \cdot \vec{\mathbf{E}}_2 \rangle = \tfrac{1}{2}(\vec{\mathbf{E}}_1 \cdot \vec{\mathbf{E}}_2^* + \vec{\mathbf{E}}_1^* \cdot \vec{\mathbf{E}}_2)$$

9.2 In Section 9.1 we considered the spatial distribution of energy for two point sources. We mentioned that for the case in which the separation $a >> \lambda$, I_{12} spatially averages to zero. Why is this true? What happens when a is much less than λ?

9.2 The largest value of $(r_1 - r_2)$ is equal to a. Thus if $\varepsilon_1 = \varepsilon_2$, $\delta = k(r_1 - r_2)$ varies from 0 to ka. If $a >> \lambda$, $\cos \delta$ and therefore I_{12} will have a great many maxima and minima and therefore average to zero over a large region of space. In contrast, if $a << \lambda$, δ varies only slightly from 0 to $ka << 2\pi$. Hence I_{12} does not average to zero. and from Eq. (9.17), I deviates little from $4I_0$. The two sources effectively behave as a single source of double the original strength.

9.4 Will we get an interference pattern in Young's Experiment (Fig. 9.8) if we replace the source slit S by a single long-filament light-bulb? What would occur if we replaced the slits S_1 and S_2 by these same bulbs?

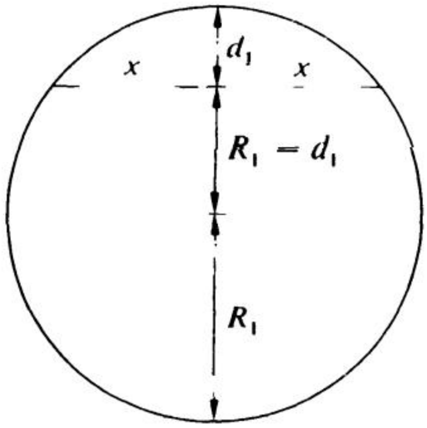
9.4 A bulb at S would produce fringes. We can imagine it as made up of a very large number of incoherent point sources. Each of these would generate an independent pattern, all of which would then overlap. Bulbs at S_1 and S_2 would be incoherent and could not generate detectable fringes.

9.26 A soap film surrounded by air has an index of refraction of 1.34. If a region of the film appears bright red ($\lambda_0 = 633$ nm) in normally reflected light, what is its minimum thickness there?

9.26 Here $1.00 < 1.34 > 1.00$, hence from Eq. (9.36) with $m = 0$, $d = (0 + \frac{1}{2})\,(633\text{ nm})/2(1.34) = 118\text{ nm}$.

9.29 Consider the circular pattern of Haidinger's fringes resulting from a film with a thickness of 2 mm and an index of refraction of 1.5. For monochromatic illumination of $\lambda_0 = 600$ nm, find the value of m for the central fringe ($\theta_t = 0$). Will it be bright or dark?

9.29 Eq. (9.37) $m = 2n_f d/\lambda_0 = 10\,000$. A minimum, therefore central dark region.



Homework

- ❑ 9.31, 9.33, 9.45, 9.46

9.31 Figure P.9.31 illustrates a setup used for testing lenses. Show that

$$d = x^2(R_2 - R_1)/2R_1R_2$$

when d_1 and d_2 are negligible in comparison with $2R_1$ and $2R_2$, respectively. (Recall the theorem from plane geometry that relates the products of the segments of intersecting chords.) Prove that the radius of the mth dark fringe is then

$$x_m = [R_1R_2m\lambda_f/(R_2 - R_1)]^{1/2}$$

How does this relate to Eq. (9.43)?

9.31 $x^2 = d_1[(R_1 - d_1) + R_1] = 2R_1d_1 - d_1^2$

Similarly $x^2 = 2R_2d_2 - d_2^2$

$$d = d_1 - d_2 = \frac{x^2}{2}\left[\frac{1}{R_1} - \frac{1}{R_2}\right], \qquad d = m\frac{\lambda_f}{2}$$

As $R_2 \to \infty$, x_m approaches Eq. (9.43).

9.33 Fringes are observed when a parallel beam of light of wavelength 500 nm is incident perpendicularly onto a wedge-shaped film with an index of refraction of 1.5. What is the angle of the wedge if the fringe separation is $\frac{1}{3}$ cm?

9.33 $\Delta x = \lambda_f/2\alpha, \qquad \alpha = \lambda_0/2n_f\Delta x$

$$\alpha = 5 \times 10^{-5}\ \text{rad} = 10.2\ \text{seconds}$$

9.45 Determine the refractive index and thickness of a film to be deposited on a glass surface ($n_g = 1.54$) such that no normally incident light of wavelength 540 nm is reflected.

9.45 $n_0 = 1 \quad n_s = n_g \quad n_1 = \sqrt{n_g}$

$$\sqrt{1.54} = 1.24$$

$$d = \frac{1}{4}\lambda_f = \frac{1}{4}\frac{\lambda_0}{n_1} = \frac{1}{4}\frac{540}{1.24}\text{ nm}$$

No relative phase shift between two waves.

9.46 A glass microscope lens having an index of 1.55 is to be coated with a magnesium fluoride film to increase the transmission of normally incident yellow light ($\lambda_0 = 550$ nm). What minimum thickness should be deposited on the lens?

9.46 The refracted wave will traverse the film twice, and there will be no relative phase shift on reflection. Hence

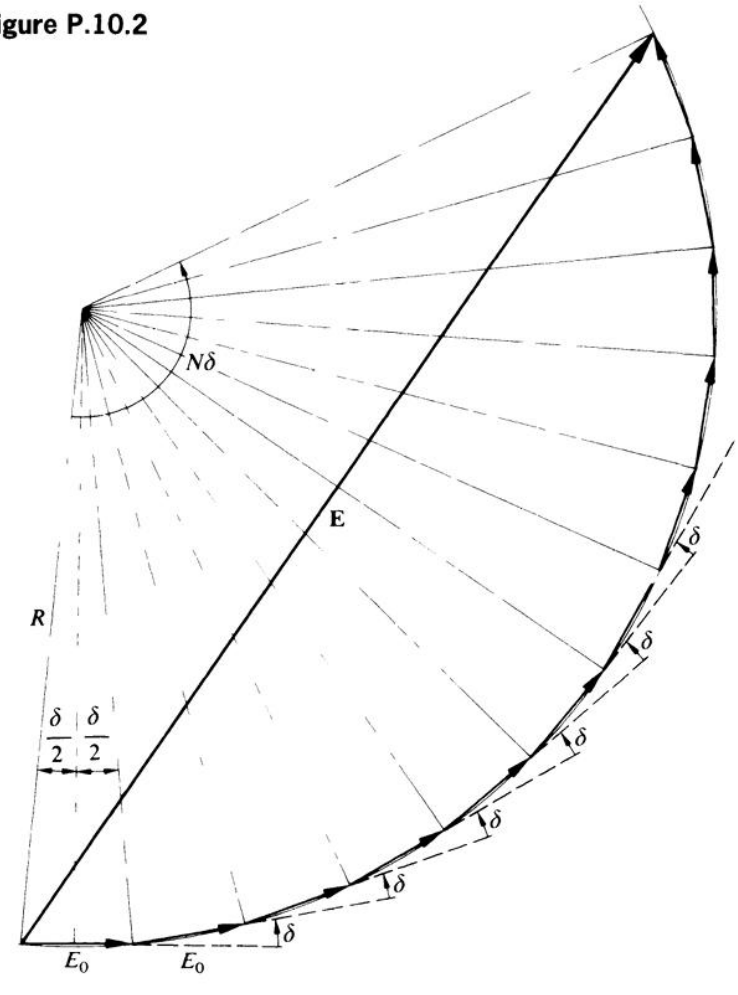
$$d = \lambda_0/4n_f = (550\text{ nm})/4(1.38) = 99.6\text{ nm}$$

Homework

- 10.2, 10.10, 10.12

10.2 Using Fig. P.10.2, derive the irradiance equation for N coherent oscillators, Eq. (10.5).

Figure P.10.2



10.2 $E_0/2 = R \sin(\delta/2)$

$$E = 2R \sin(N\delta/2) \quad \text{chord length}$$

$$E = [E_0 \sin(N\delta/2)]/\sin(\delta/2)$$

$$I = E^2$$

10.10 Show that for a double-slit Fraunhofer pattern, if $a = mb$, the number of bright fringes (or parts thereof) within the central diffraction maximum will be equal to $2m$.

10.10 $\alpha = \dfrac{ka}{2}\sin\theta, \qquad \beta = \dfrac{kb}{2}\sin\theta$

$$a = mb,\ \alpha = m\beta,\ \alpha = m2\pi$$

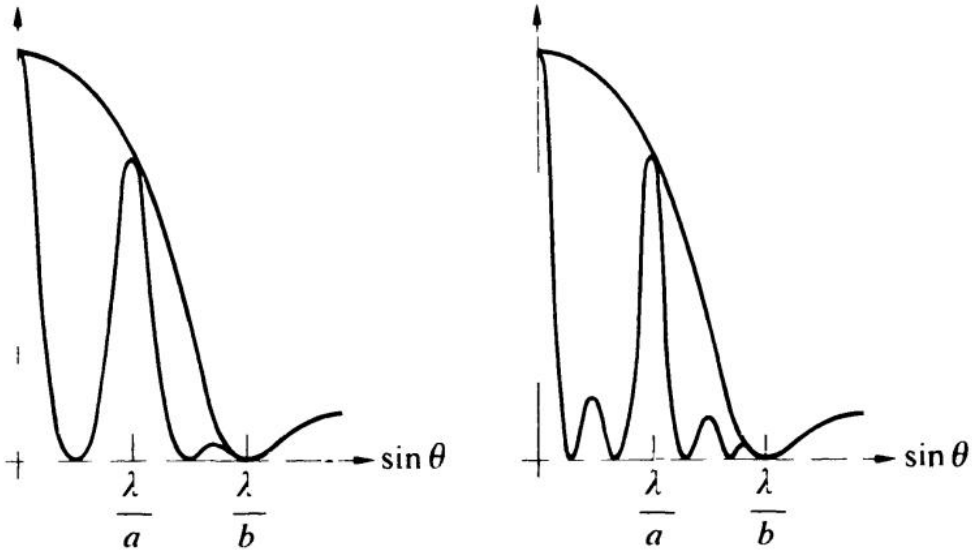
$$N = \text{number of fringes} = a/\pi = m2\pi/\pi = 2m$$

10.12 What is the relative irradiance of the subsidiary maxima in a three-slit Fraunhofer diffraction pattern? Draw a graph of the irradiance distribution, when $a = 2b$, for two and then three slits.

10.12 $\alpha = 3\pi/2N = \pi/2$ [10.34]

$$I(\theta) = \frac{I(0)}{N^2}\left(\frac{\sin\beta}{\beta}\right)^2 \qquad \text{from Eq. (10.35)}$$

and $I/I(0) \approx \frac{1}{9}$.



Homework

- 10.32, 10.35, 10.37

10.32 White light falls normally on a transmission grating that contains 1000 lines per centimeter. At what angle will red light ($\lambda_0 = 650$ nm) emerge in the first-order spectrum?

10.32 From Eq. (10.32), where $a = 1/(1000 \text{ lines per cm}) = 0.001$ cm per line (center to center), $\sin \theta_m = 1(650 \times 10^{-9} \text{ m})/(0.001 \times 10^{-2} \text{ m}) = 6.5 \times 10^{-2}$ and $\theta_1 = 3.73°$.

10.35 Light having a frequency of 4.0×10^{14} Hz is incident on a grating formed with 10 000 lines per centimeter. What is the highest-order spectrum that can be seen with this device? Explain.

10.35 The largest value of m in Eq. (10.32) occurs when the sine function is equal to one, making the left side of the equation as large as possible, then $m = a/\lambda = (1/10 \times 10^5)/(3.0 \times 10^8 \text{ m/s} \div 4.0 \times 10^{14} \text{ Hz}) = 1.3$, and only the first-order spectrum is visible.

10.37 Prove that the equation

$$a(\sin\theta_m - \sin\theta_i) = m\lambda \qquad [10.61]$$

when applied to a transmission grating, is independent of the refractive index.

10.37 $\sin\theta_i = n\sin\theta_n$

Optical path length difference $= m\lambda$

$a\sin\theta_m - na\sin\theta_n = m\lambda$

$a(\sin\theta_m - \sin\theta_i) = m\lambda$